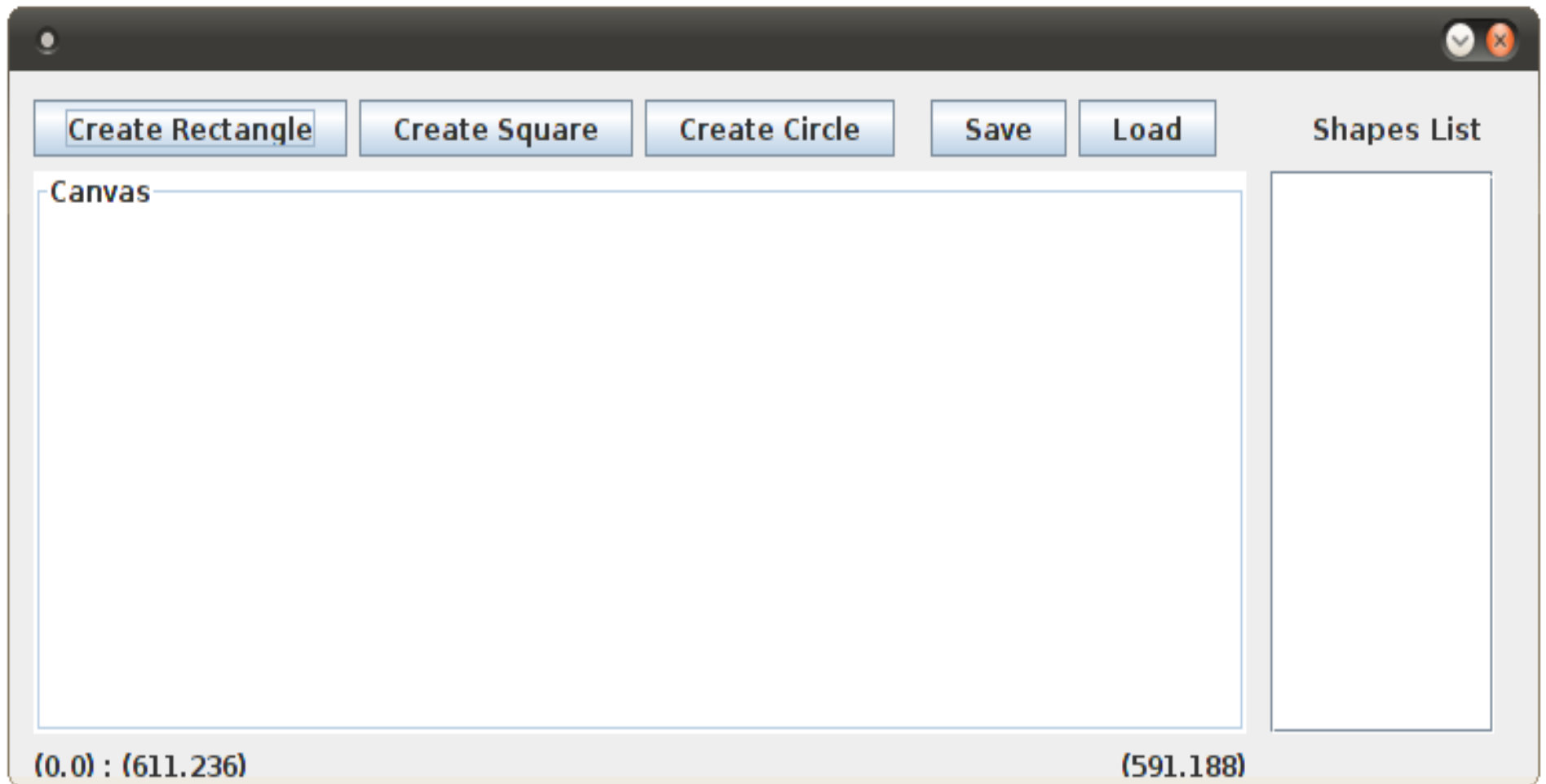
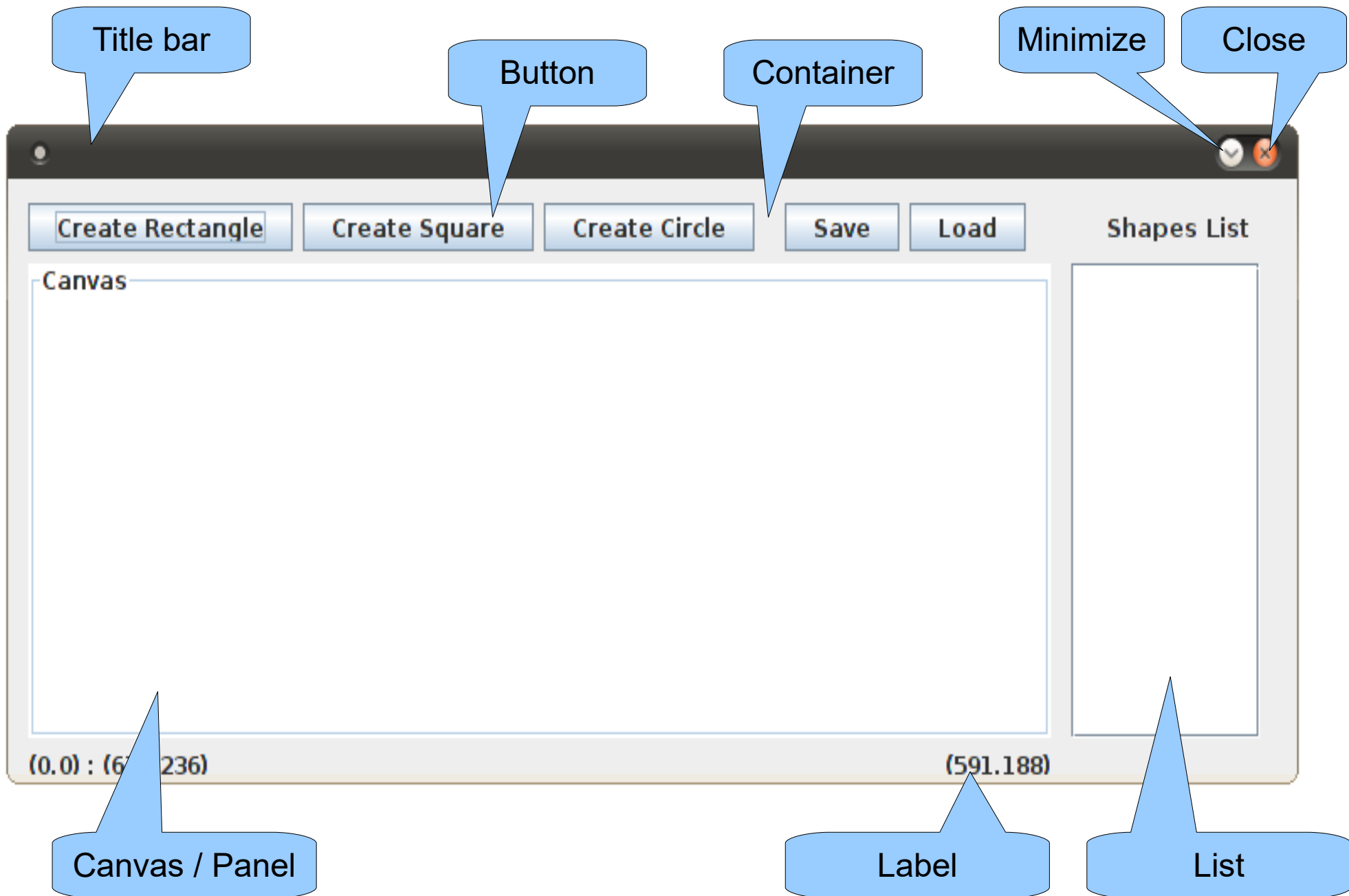
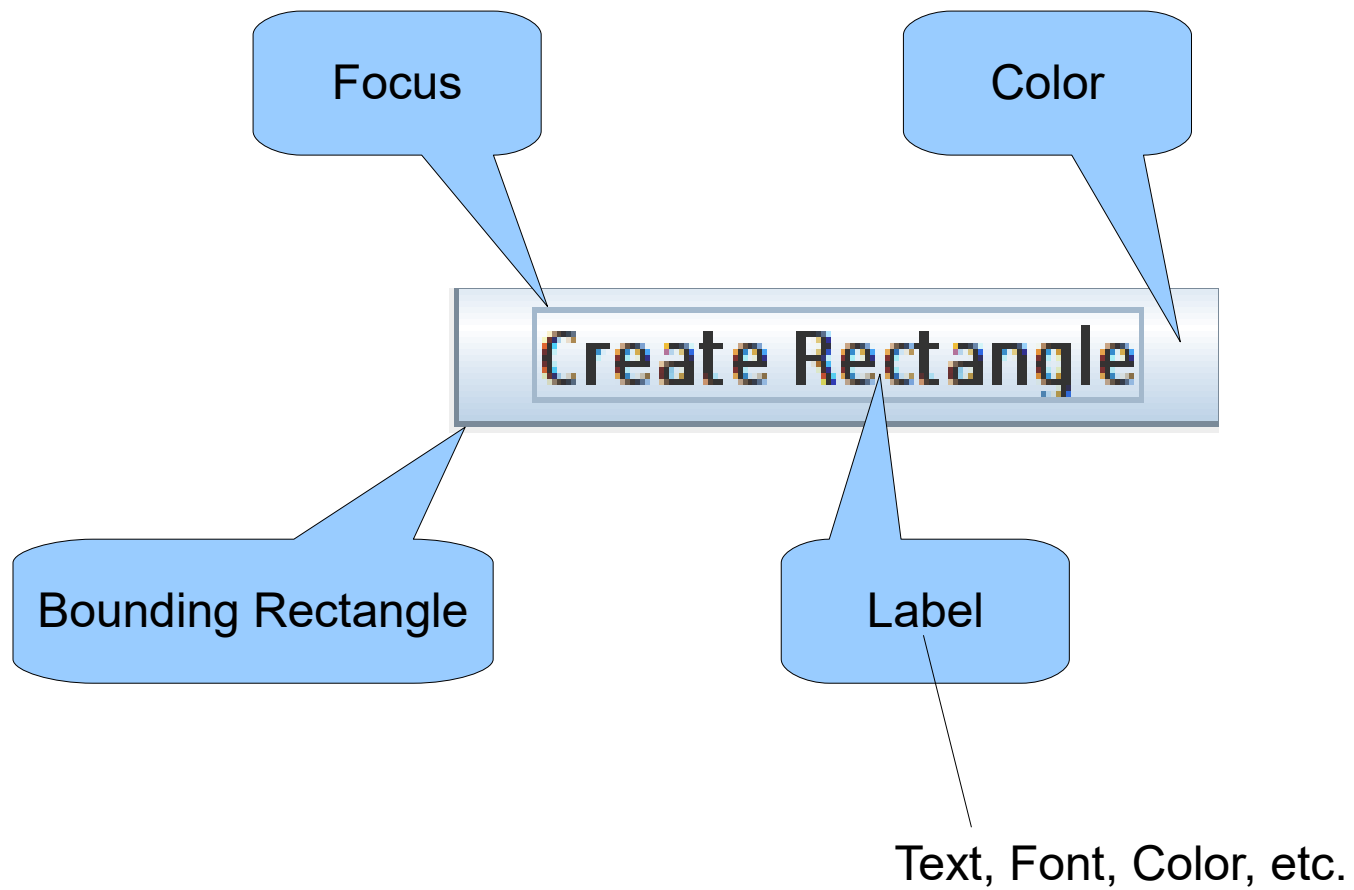
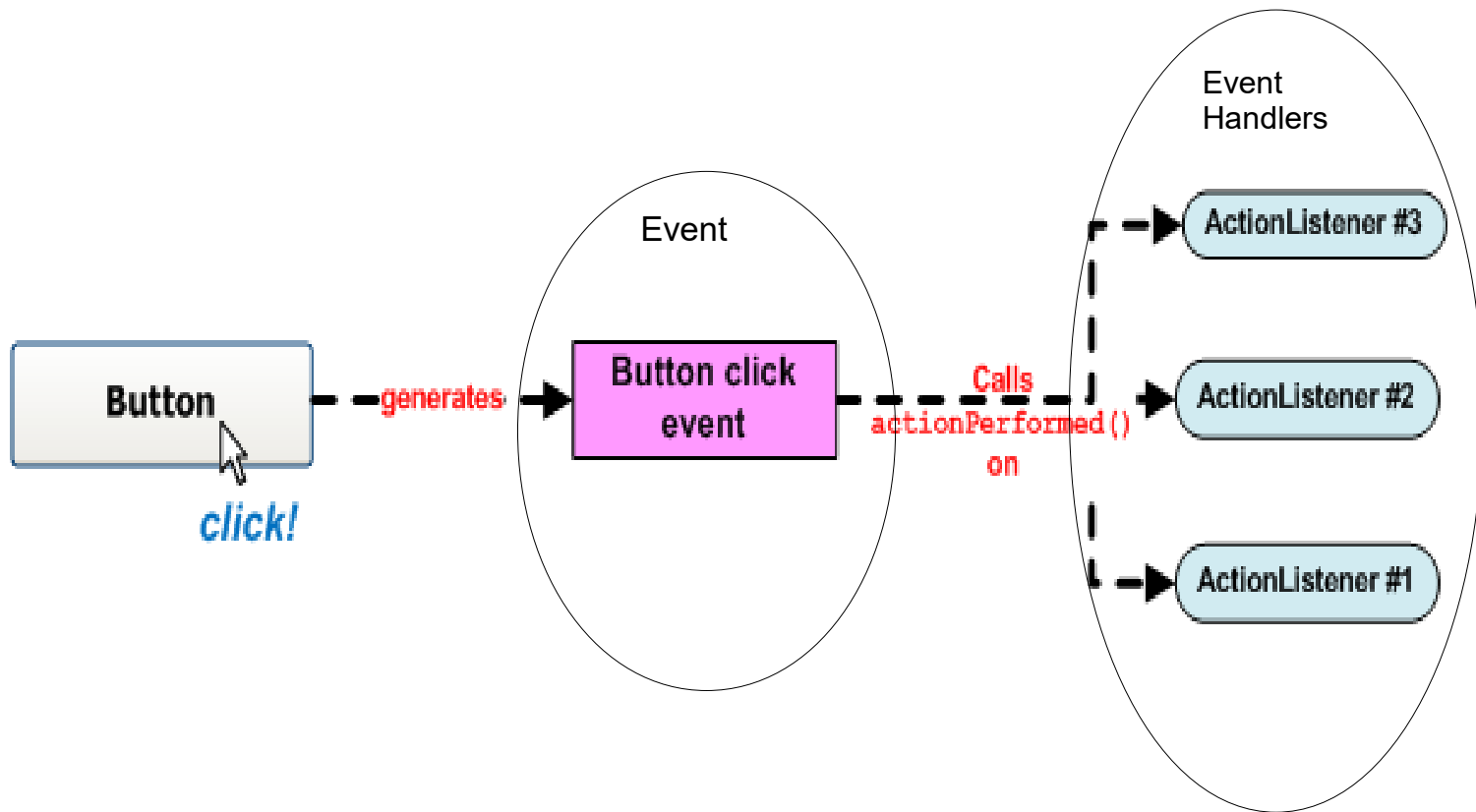


GUI Programming



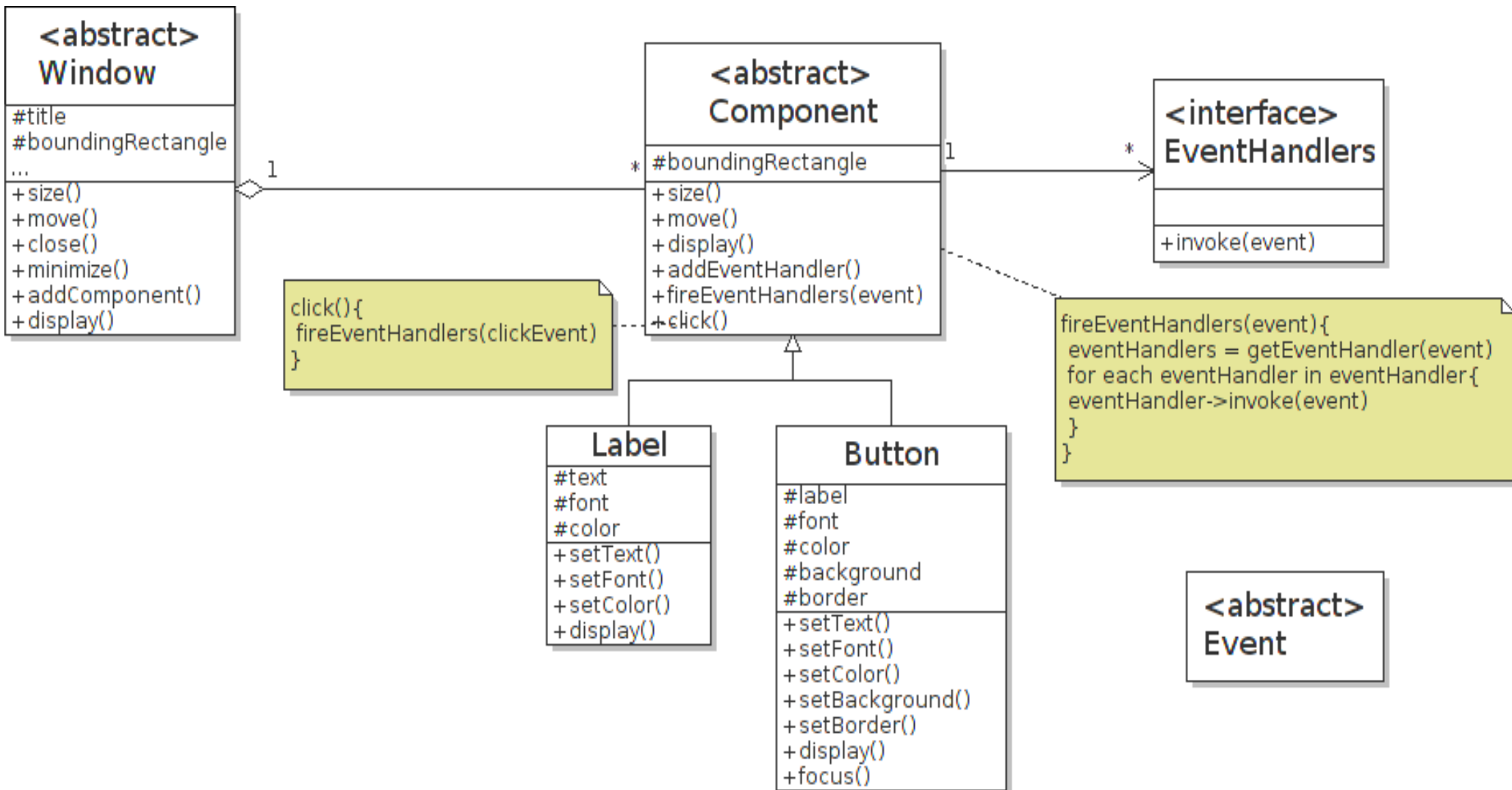




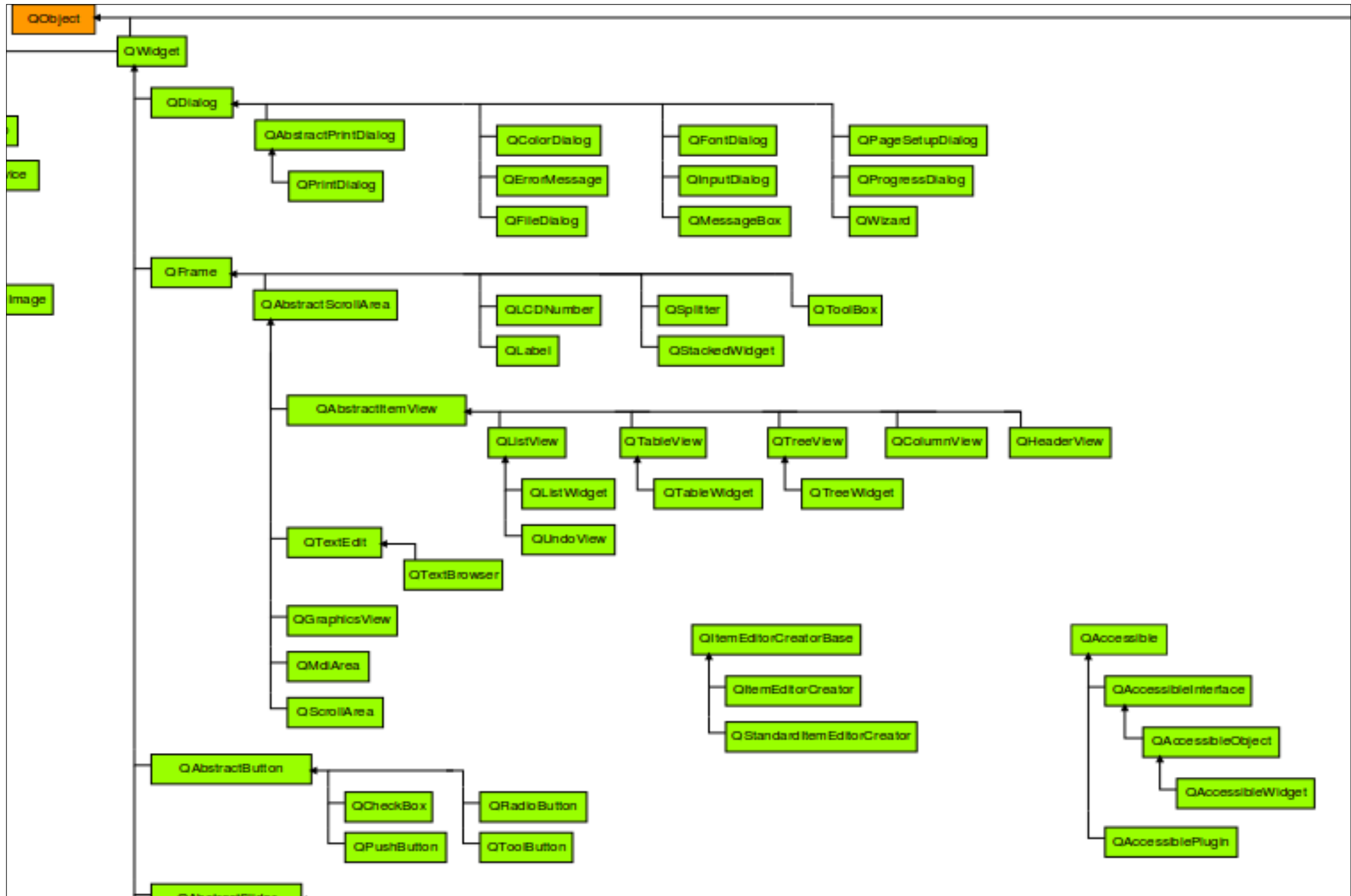


Source: <https://www.clear.rice.edu/comp310/JavaResources/GUI/>

Abstract Window Framework



QT

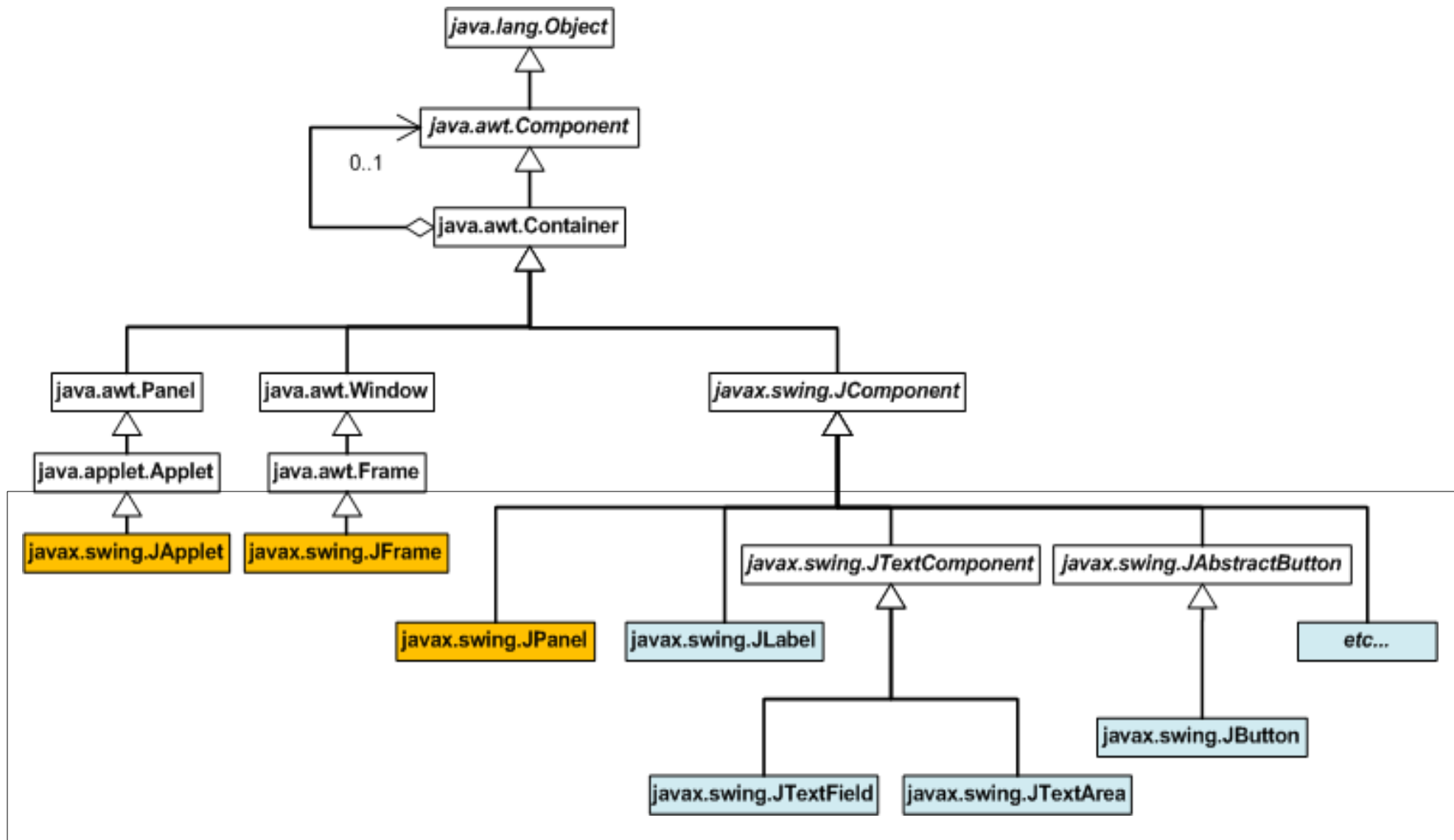


QT-based Implementation

```
class GraphicsEditor : public QWidget{
protected:
    QPushButton* buttonRectangle;
    QPushButton* buttonSquare;
    QPushButton* buttonCircle;
    QLabel* label;
    QListWidget* list;
    QGraphicsView* canvas;

public:
    GraphicsEditor();
    virtual ~GraphicsEditor();
};
```


Java Swing



Swing-based Implementation

```
class GraphicsEditor extends JFrame{  
  
    protected JButton buttonRectangle;  
    protected JButton buttonSquare;  
    protected JButton buttonCircle;  
    protected JLabel label;  
    protected JList list;  
    protected JPanel canvas;  
  
    public GraphicsEditor(){  
        ...  
    }  
}
```